

Ch 5 Review: Continuous Random Variables

Remember a continuous random variable X is defined by its **cumulative distribution function**

$$F_X(t) := P(X \leq t)$$

or equivalently defined by its **density function**

$$f_X(t) := \frac{d}{dt} P(X \leq t) \longleftrightarrow F_X(a) = \int_{-\infty}^a f_X(t) dt$$

What do we remember about these objects?

$$f_X(t) \neq P(\text{something})$$

$$f_X(t) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(t) dt = 1$$

$$\lim_{t \rightarrow \infty} F_X(t) = P(X \leq \infty) = 1$$

$$\lim_{t \rightarrow -\infty} F_X(t) = P(X \leq -\infty) = 0$$

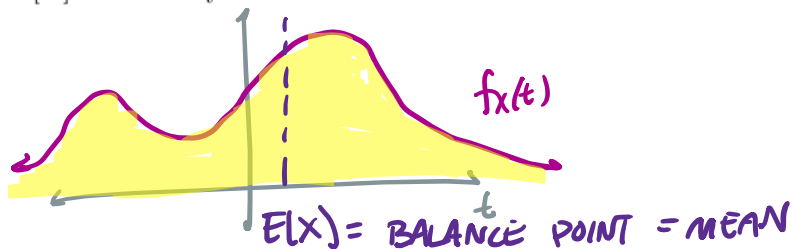
$F_X(t)$ is increasing

Besides the probability distribution of X , we can also calculate expectations of (functions of) X :

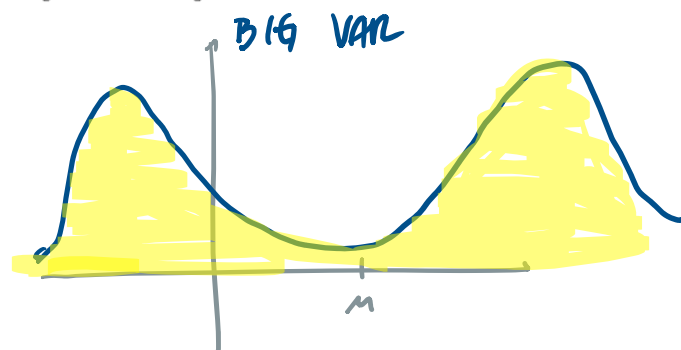
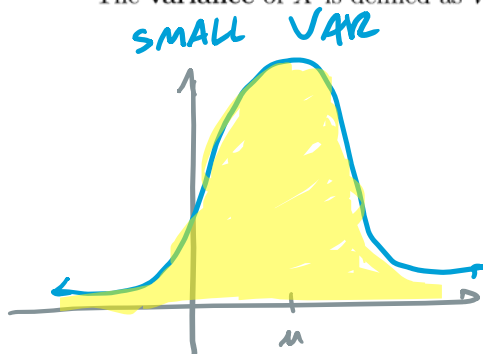
$$E[g(X)] := \int_{-\infty}^{\infty} g(t) f_X(t) dt$$

$$\sum_a g(a) \cdot p_X(a)$$

The value $E[X]$ is commonly referred to as the **mean** of X



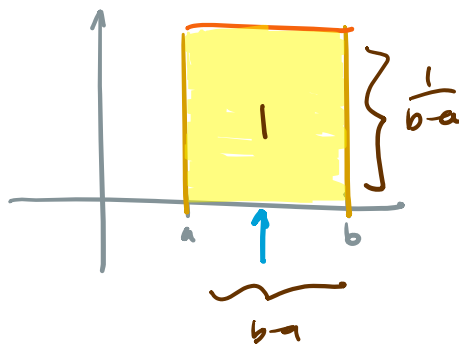
The **variance** of X is defined as $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$



A few famous continuous random variables you should be familiar with:

I. Uniform

Density	Mean	Variance
$f_X(t) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$



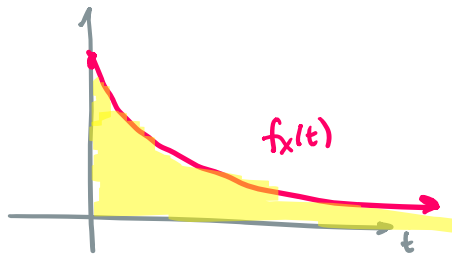
$$\begin{aligned}
 E[X] &= \int_{-\infty}^{\infty} t \cdot f_X(t) dt \\
 &= \int_{-\infty}^a t \cdot 0 dt + \int_a^b t \cdot \frac{1}{b-a} dt + \int_b^{\infty} t \cdot 0 dt \\
 &= \frac{1}{b-a} \cdot \int_a^b t dt = \frac{1}{b-a} \cdot \left. \frac{t^2}{2} \right|_a^b \\
 &= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}
 \end{aligned}$$

$$\begin{aligned}
 E[X^2] &= \int_{-\infty}^{\infty} t^2 f_X(t) dt = \int_a^b t^2 \cdot \frac{1}{b-a} dt = \frac{1}{b-a} \cdot \left. \frac{t^3}{3} \right|_a^b \\
 &= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{b^2 + ab + a^2}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - (E[X])^2 = \frac{b^2 + ab + a^2}{3} - \left(\frac{b+a}{2} \right)^2 \\
 &= \frac{4b^2 + 4ab + 4a^2}{12} - \frac{3b^2 + 6ab + 3a^2}{12} = \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}
 \end{aligned}$$

II. Exponential

Density	Mean	Variance
$f_X(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & \text{else} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$



$$E(X) = \int_{-\infty}^{\infty} t \cdot f_X(t) dt = \int_0^{\infty} t \cdot \lambda e^{-\lambda t} dt$$

$$= -t e^{-\lambda t} - \frac{1}{\lambda} e^{-\lambda t} \Big|_0^{\infty}$$

$$= (0 - 0) - (0 - \frac{1}{\lambda} \cdot 1)$$

$$= \frac{1}{\lambda}$$

Tabular:

+	t	$\lambda e^{-\lambda t}$
-	1	$-e^{-\lambda t}$
0		$\frac{1}{\lambda} e^{-\lambda t}$

$$E(X^2) = \int_{-\infty}^{\infty} t^2 f_X(t) dt = \int_0^{\infty} t^2 \cdot \lambda e^{-\lambda t} dt$$

$$= -t^2 e^{-\lambda t} - \frac{2t}{\lambda} e^{-\lambda t} - \frac{2}{\lambda^2} e^{-\lambda t} \Big|_0^{\infty}$$

$$= (0 - 0 - 0) - (0 - 0 - \frac{2}{\lambda^2} \cdot 1) = \frac{2}{\lambda^2}$$

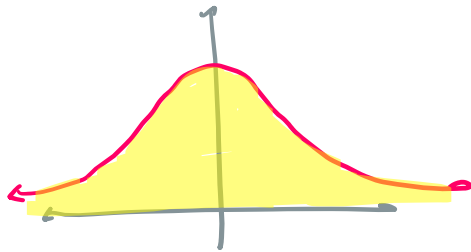
+	t ²	$\lambda e^{-\lambda t}$
-	2t	$-e^{-\lambda t}$
+	2	$\frac{1}{\lambda} e^{-\lambda t}$
0		$-\frac{1}{\lambda^2} e^{-\lambda t}$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

III. Standard Normal

Density	Mean	Variance
$f_Z(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$	0	1

The cumulative distribution function for the standard normal is denoted $\Phi(x)$



$$F_Z(x) = P(Z \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \Phi(x)$$

$$\begin{aligned} E[Z] &= \int_{-\infty}^{\infty} t \cdot f_X(t) dt \\ &= \int_{-\infty}^{\infty} t \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \\ &= -\frac{1}{\sqrt{2\pi}} e^{-t^2/2} \Big|_{-\infty}^{\infty} \\ &= 0 - 0 = 0 \end{aligned}$$

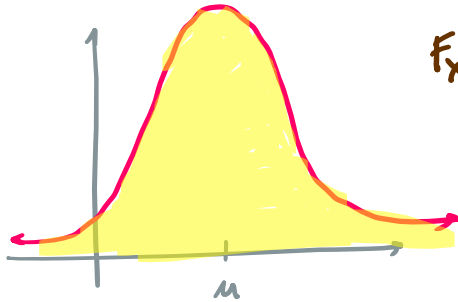
$$\begin{aligned} \left(\int_{-\infty}^{\infty} e^{-t^2/2} dt \right)^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2/2} \cdot e^{-y^2/2} \cdot dx dy \\ &= \iint_{\mathbb{R}^2} e^{-\frac{(x^2+y^2)}{2}} dA = \int_0^{2\pi} \int_0^{\infty} e^{-\frac{r^2}{2}} \cdot r dr d\theta \\ &= \int_0^{2\pi} d\theta \cdot \int_0^{\infty} r e^{-r^2/2} dr = 2\pi \cdot \left[-e^{-r^2/2} \right]_0^{\infty} = 2\pi \cdot 1 = 2\pi \end{aligned}$$

$$E[Z^2] = \int_{-\infty}^{\infty} t^2 f_X(t) dt = \int_{-\infty}^{\infty} t^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

$$\begin{aligned} u &= t & dv &= t e^{-t^2/2} dt \\ \frac{du}{dt} &= 1 & v &= -e^{-t^2/2} \end{aligned}$$

$$\begin{aligned} &= \frac{t}{\sqrt{2\pi}} \cdot (-e^{-t^2/2}) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -e^{-t^2/2} \cdot \frac{1}{\sqrt{2\pi}} dt \\ &= (0 - 0) + \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt}_1 = 1 \end{aligned}$$

If $Z = \mathcal{N}(0, 1)$ is the standard normal, then $X := \sigma Z + \mu$ is a normal random variable with mean μ and variance σ^2 . We denote this as $X = \mathcal{N}(\mu, \sigma^2)$. What's the density and cdf of X ?

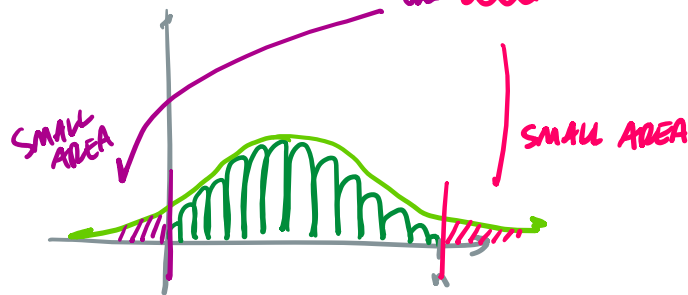


$$F_X(t) = P(X \leq t) = P(\sigma Z + \mu \leq t) \\ = P(Z \leq \frac{t - \mu}{\sigma}) = \Phi\left(\frac{t - \mu}{\sigma}\right)$$

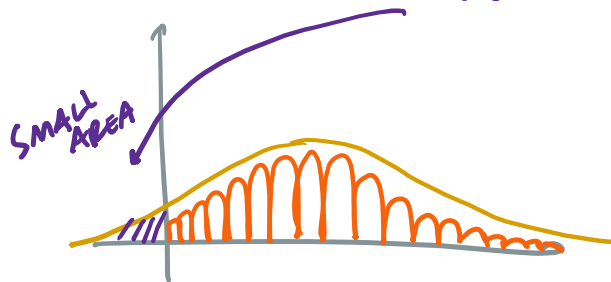
$$f_X(t) = \frac{d}{dt} F_X(t) = \frac{d}{dt} \Phi\left(\frac{t - \mu}{\sigma}\right) \\ = \Phi'\left(\frac{t - \mu}{\sigma}\right) \cdot \frac{1}{\sigma} = \frac{1}{\sqrt{\pi}} e^{-\frac{(\frac{t - \mu}{\sigma})^2}{2}} \cdot \frac{1}{\sigma} \\ = \frac{1}{\sqrt{\pi \sigma^2}} e^{-\frac{(t - \mu)^2}{2\sigma^2}}$$

It is common in probability and statistics to assume a distribution is normal

$$\text{Bin}(n, p) \approx \mathcal{N}(np, np(1 - p)) \text{ when } \min(np, n(1 - p)) \text{ is large}$$



$$\text{Pois}(\lambda) \approx \mathcal{N}(\lambda, \lambda) \text{ when } \lambda \text{ is large}$$



Given the inconvenient nature of integrating in $\Phi(x)$, why does the normal distribution show up so much?

WE WILL SEE IN CT1 8!